Refinements to the SAE–Thompson Sampling Framework for Active Email Outreach

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1 Introduction

Žid et al. [2025] frame cold-start recipient selection for a new email campaign as an iterative multi-armed bandit: each arm is a recipient, and Thompson Sampling (TS) parameters are derived from a shallow autoencoder (SAE) trained on historical recipient–template opens, avoiding retraining during the active-learning phase. Several components of that model — a fixed exploration/exploitation coefficient, an unweighted historic open rate, a linear aggregation for the current-state confidence term, training on binary labels only, and a shallow autoencoder architecture — were fixed by the original authors for simplicity. We revisit each of these choices and propose concrete, mathematically grounded refinements: dynamic scheduling of the blend coefficient, a recency-weighted historic rate, a nonlinear forward-pass confidence term, a variance-based influence score, a factored score function, two ways of exposing the autoencoder to previously unused time-to-open (TTO) signal, and a deep, nonlinear autoencoder. We report experimental results for each proposal against a reproduced baseline and outline directions for further work.

2 Recap of the Original Model

Setup. We observe $X \in \{0,1\}^{n\times m}$, where n is the number of templates, m the number of recipients, and $X_{i,j} = 1$ if recipient j opened template i. For a new template $n+1$, sent over b batches within a deadline T, the model selects at each batch a small subset of recipients so as to maximize eventual openers reached early.

Shallow autoencoder. A SAE with encoder/decoder $E, D \in \mathbb{R}^{m\times d}$ is trained as

$$\min_{E,D} \ell\big(X,\ \sigma(XB_{E,D})\big), \qquad B_{E,D} = ED^\top - \operatorname{diag}\big([E \odot D]\mathbf{1}_m\big),$$

with ℓ the element-wise binary cross-entropy and the diagonal constraint ruling out trivial self-prediction. We write $f_{\mathrm{SAE}}(x) = \sigma(x^\top B_{E,D})$ for the trained forward pass and $\Sigma = \sigma(B_{E,D})$; entry $\Sigma_{i,j}$ is the probability of recipient j opening a template given that only recipient i has opened it — a measure of i's influence on j.

Arm score. The bandit score for recipient j at time t is

$$s_j(t) = \alpha\, \phi_j p_j + (1-\alpha) f_j(t), \tag{1}$$

$$\phi_j = \frac{1}{n}\sum_{i=1}^{n} X_{i,j}, \qquad p_j = \frac{1}{m-1}\sum_{i\neq j} \Sigma_{j,i}, \qquad f_j(t) = \bar{x}(t)^\top \Sigma_{:j}, \tag{2}$$

where ϕ_j is recipient j's historic open rate, p_j is a general influence score, $f_j(t)$ is a confidence score specific to template $n+1$ given the observed (normalized) state $\bar{x}(t)$, and $\alpha \in [0,1]$ trades off the two. TS parameters are then $\alpha_j(t) = Gs_j(t)$, $\beta_j(t) = G(1-s_j(t))$ for a confidence modifier G.

Two orthogonal readings of Eq. (1) motivate our refinements: $\phi_j p_j$ vs. $f_j(t)$ can be read as *exploration vs. exploitation*, or equivalently as *historic vs. current* data.

3 Proposed Refinements

3.1 Dynamic α-Scheduling

A fixed α ignores that the value of exploration should shrink as the campaign progresses, echoing the broader practice of decaying exploration coefficients over time in the bandit and reinforcement-learning literature [Auer et al., 2002]. We schedule $\alpha(t)$ against two natural progress variables: $\mu = N(t)/N$, the fraction of the sending budget already used, and $\tilde{o} = o(t)/m$, the fraction of recipients who have already opened, with $o(t) = x(t)^\top \mathbf{1}$.

Send-volume-based (function of μ). A **linear** schedule from l to r, in the spirit of the linearly annealed ε-greedy exploration rate used to train DQN agents [Mnih et al., 2015],

$$\alpha(\mu) = l(1-\mu) + r\mu, \tag{3}$$

and a **geometric** (log-linear) schedule, analogous to the geometric cooling schedules of simulated annealing [Mahdi et al., 2017],

$$\alpha(\mu) = l^{1-\mu}\, r^{\mu}. \tag{4}$$

Open-rate-based (function of $\tilde{o}$). Since $f_j(t)$ is itself a point estimate that becomes more reliable as more opens are observed, we also consider a **confidence** schedule, hyperbolic in $\tilde{o}$ — paralleling how the confidence radius of UCB-style bandit policies shrinks as observations accumulate [Auer et al., 2002]:

$$\alpha(\tilde{o}) = \frac{\kappa}{\tilde{o}+\kappa}, \tag{5}$$

with dampening factor κ (so $\alpha(\kappa) = \frac{1}{2}$). All three reduce to the constant baseline at their degenerate limits ($l = r$, or $\kappa \to \infty$). A schedule jointly conditioned on both μ and $\tilde{o}$ is conceptually natural but is left to future work (Section 5) to avoid an unconstrained hyperparameter count.

3.2 Recency-Weighted Historic Rate ϕ_j

The original ϕ_j weights every past template equally, ignoring drift in recipient behavior over time — a limitation well studied in time-aware collaborative filtering [Ding and Li, 2005, Koren, 2009]. Assuming templates $X_{1:}, \ldots, X_{n:}$ were sent at uniform intervals, we assign template i an exponential weight, following the time-weighting scheme of Ding and Li [2005],

$$\omega_i = 2^{-d_i/h}, \qquad d_i = \frac{n-i}{n}, \tag{6}$$

with half-life $h > 0$, and redefine

$$\phi_j = \frac{\omega^\top X_{:j}}{\sum_i \omega_i}. \tag{7}$$

Normalizing by $\sum_i \omega_i$ keeps $\phi_j \in [0,1]$. As $h \to \infty$ every $\omega_i \to 1$, recovering the original unweighted average as a special case.

3.3 Forward-Pass Definition of $f_j(t)$

$f_j(t) = \bar{x}(t)^\top \Sigma_{:j}$ is a linear combination of individual influences $\Sigma_{i,j}$ and therefore captures only the *average* effect of openers, never their joint interaction. We instead pass the full current state through the trained network, in the spirit of nonlinear autoencoder-based collaborative-filtering predictors such as AutoRec [Sedhain et al., 2015]:

$$f_j(t) = \sigma\big(x(t)^\top B_{E,D}\big) e_j = f_{\mathrm{SAE}}(x(t))\, e_j. \tag{8}$$

Because σ is nonlinear, $\sigma(x(t)^\top B_{E,D}) \neq x(t)^\top \sigma(B_{E,D})$ in general (equality holds only for one-hot $x(t)$), so Eq. (8) is a genuinely different, jointly-conditioned probability estimate rather than a relabeling of Eq. (2).

3.4 Variance-Based Definition of p_j

p_j is meant to reward recipients whose opening is *informative* about others, in the sense of informativeness used throughout active learning. The original definition, $p_j = \frac{1}{m-1}\sum_{i\neq j} \Sigma_{j,i}$, rewards recipients that raise the *average level* of other recipients' f-scores — which is not the same as reducing uncertainty about them. We instead reward maximal *spread* induced across other recipients' scores, following variance-based exploration bonuses in the multi-armed bandit literature [Audibert et al., 2009]:

$$p_j = \frac{4}{m-1}\sum_{i\neq j}\big(\Sigma_{j,i} - \bar{\Sigma}_{j:}\big)^2, \qquad \bar{\Sigma}_{j:} = \frac{1}{m-1}\sum_{i\neq j}\Sigma_{j,i}. \tag{9}$$

This is the variance of $\Sigma_{j,:}$ (excluding j), scaled by 4 so that the maximum possible variance of a Bernoulli variable (0.25) maps to 1, keeping $p_j \in [0,1]$.

3.5 Factored Score $s_j(t) = \pi_j(t)\, u_j(t)$

The linear combination in Eq. (1) conflates the two orthogonal axes noted in Section 2 — it is not clear, for instance, why exploration should draw solely on historic data while exploitation draws solely on current data. We instead factor the score into a *probability-of-opening* term and a *utility-of-opening* term, each independently schedulable:

$$\pi_j(t) = \alpha(t)\phi_j + (1-\alpha(t)) f_j(t), \qquad u_j(t) = \beta(t) p_j + (1-\beta(t))\cdot 1, \tag{10}$$

$$s_j(t) = \pi_j(t)\, u_j(t) = \Big(\alpha(t)\phi_j + [1-\alpha(t)] f_j(t)\Big)\Big(\beta(t) p_j + [1-\beta(t)]\Big). \tag{11}$$

Here $\pi_j(t)$ blends historic and current opening-probability estimates, while $u_j(t)$ blends the direct value of an open (the constant 1) with its indirect information value p_j. This factorization reuses every primitive introduced above unchanged, so refinements from Sections 3.1–3.4 can be dropped into $\pi_j(t)$ and $u_j(t)$ directly — we use a confidence schedule (Eq. (5)) for $\alpha(t)$ and a send-volume schedule (Eq. (3)/(4)) for $\beta(t)$.

3.6 Incorporating Time-to-Open (TTO)

The SAE is trained purely on binary opens, discarding *how fast* a recipient opened — signal shown to carry predictive value for engagement [Sinha et al., 2018] and that matters given the short operational window $(T/b,\, T)$. Let $\Delta_{i,j} \in \mathbb{R}_{\geq 0} \cup \{+\infty\}$ be the observed time-to-open (with $\Delta_{i,j} = +\infty$ for non-openers, so $X_{i,j} = \mathbb{1}[\Delta_{i,j} < \infty]$), and define a decayed open label

$$Y_{i,j} = 2^{-\Delta_{i,j}/h_\delta}, \qquad 2^{-\infty} := 0, \tag{12}$$

with TTO half-life h_δ; as $h_\delta \to \infty$, $Y_{i,j} \to X_{i,j}$ for every opener.

Binary-to-TTO autoencoder. We keep the SAE's binary input but replace the reconstruction target:

$$\min_{E,D} \ \ell\big(Y,\ \sigma(XB_{E,D})\big). \tag{13}$$

Since the input space is unchanged, the forward-pass definitions of p_j and $f_j(t)$ (Eqs. (8)–(9)) carry over mechanically; only their interpretation shifts — $\Sigma_{j,i}$ now blends *whether* i opens with *how fast*.

TTO cutoff. Training rows of X reflect the *eventual* open pattern, while active-learning queries $x(t)$ are necessarily censored at $t < T$. We instead censor the training input itself, fixing a cutoff δ_c and defining $C_{i,j} = \mathbb{1}[\Delta_{i,j} \leq \delta_c]$, then training

$$\min_{E,D} \ \ell\big(X,\ \sigma(CB_{E,D})\big). \tag{14}$$

C retains only opens fast enough to plausibly be observed within an active operational window and treats slower opens as non-opens during training — matching the censoring recipients are actually subject to during active learning. As $\delta_c \to \infty$, $C \to X$, recovering the original model.

3.7 Deep Autoencoder

$B_{E,D}$ in Eq. (2) is a single fixed $m \times m$ matrix, so the SAE can only capture linear recipient–recipient relationships. Since p_j and $f_j(t)$ depend only on the ability to query f_{SAE}, not on any explicit property of $B_{E,D}$, we can swap in a deeper, nonlinear network without changing either definition. We use a two-layer encoder/decoder with a ReLU nonlinearity, following the deep collaborative-filtering autoencoders of Kuchaiev and Ginsburg [2017], which were shown to generalize better than their shallow counterparts,

$$\text{Encoder: } \mathrm{Lin}(m, 2d) \to \mathrm{ReLU} \to \mathrm{Lin}(2d, d), \qquad \text{Decoder: } \mathrm{Lin}(d, 2d) \to \mathrm{ReLU} \to \mathrm{Lin}(2d, m). \tag{15}$$

giving $f_{\text{DAE}}(x) = (\sigma \circ \text{Decoder} \circ \text{Encoder})(x)$, optionally with dropout [Srivastava et al., 2014] and bottleneck layer normalization. A deep network has no single weight matrix whose diagonal can be constrained to zero; we substitute a denoising objective [Vincent et al., 2008], randomly masking input entries and requiring reconstruction of the full unmasked X — as in collaborative denoising autoencoders for top-N recommendation [Wu et al., 2016] — forcing every prediction to depend on other recipients rather than a self-shortcut. Training becomes $\min_\theta \ell(X, f_{\text{DAE}}(X;\theta))$, and p_j, $f_j(t)$ retain their Sections 3.3–3.4 forms with f_{SAE} replaced by f_{DAE}.

4 Experiments

4.1 Methodology

For each variant we grid-search its hyperparameters and select the configuration maximizing validation-set Recall-AUC, AUC $= \int_0^1 \text{Recall}(\tau)\, d\tau$, where Recall($\tau$) is recall at sending fraction τ. We report test-set Recall@5/15/25/35% and Recall-AUC, mean ± std over repeated simulations. Before testing modifications we reproduced the original model on our pipeline; at the operating points reported by Žid et al. [2025] (25/50/75%) our reproduction matches the published point estimates exactly (0.923, 0.975, 0.989), with larger variance attributable to fewer simulation runs.

4.2 Results

Table 1: Test-set recall at four sending fractions and Recall-AUC (mean ± std) for each proposed refinement against the reproduced baseline. Hyperparameters were selected by grid search on the validation set.

Variant	Selected hyperparameters	Recall@5%	Recall@15%	Recall@25%	Recall@35%	AUC
Baseline	—	0.385 ± 0.076	0.821 ± 0.028	0.923 ± 0.014	0.958 ± 0.011	0.894 ± 0.012
Linear α-schedule	$l = 0.1,\ r = 0.05$	0.383 ± 0.073	0.823 ± 0.023	0.927 ± 0.012	0.960 ± 0.010	0.894 ± 0.011
Geometric α-schedule	$l = 0.3,\ r = 0.05$	0.377 ± 0.050	0.823 ± 0.017	0.927 ± 0.012	0.959 ± 0.010	0.894 ± 0.009
Confidence α-schedule	$\kappa = 0.003$	0.341 ± 0.031	0.818 ± 0.022	0.926 ± 0.014	0.960 ± 0.010	0.891 ± 0.009
Recency-weighted ϕ_j	$h = 0.3$	0.447 ± 0.041	0.837 ± 0.018	0.928 ± 0.014	0.960 ± 0.011	0.903 ± 0.009
Forward-pass $f_j(t)$	—	0.375 ± 0.049	0.841 ± 0.029	0.937 ± 0.018	0.966 ± 0.010	0.901 ± 0.011
Factored $s_j(t)$	$\alpha(t)$: conf., $\kappa = 0.005$; $\beta(t)$: geom., $l = 0.1,\ r = 0.05$	0.425 ± 0.026	0.821 ± 0.017	0.922 ± 0.013	0.959 ± 0.010	0.900 ± 0.008
Variance-based p_j	—	0.479 ± 0.034	0.837 ± 0.017	0.926 ± 0.013	0.960 ± 0.010	0.905 ± 0.009
Binary-to-TTO autoencoder	$h_\delta = 11520$	0.331 ± 0.051	0.719 ± 0.040	0.896 ± 0.015	0.955 ± 0.008	0.877 ± 0.011
TTO cutoff	$\delta_c = 720$	0.390 ± 0.038	0.824 ± 0.016	0.924 ± 0.012	0.960 ± 0.010	0.895 ± 0.008
Deep autoencoder	$d = 16$	0.404 ± 0.086	0.829 ± 0.052	0.931 ± 0.019	0.963 ± 0.011	0.900 ± 0.017

4.3 Discussion

Gains are largest at low sending fractions and shrink as τ grows, since baseline recall already exceeds 0.92 by $\tau = 25\%$. The **variance-based** p_j (0.479 at 5%), **recency-weighted** ϕ_j (0.447), and **factored** $s_j(t)$ (0.425) give the largest low-fraction gains and some of the highest AUC (0.905, 0.903, 0.900 vs. 0.894 baseline). The **forward-pass** $f_j(t)$ shows no gain at 5% but a consistent uplift from 15% onward, suggesting nonlinear aggregation only helps once significant number of recipients have opened. The α-scheduling variants are largely neutral, and the **Binary-to-TTO autoencoder** underperforms across all quantiles. **TTO cutoff** gives a small, consistent uplift at no interpretive cost, while the **deep autoencoder** improves recall modestly but with markedly higher variance, suggesting the added capacity is not yet well-regularized. All results test one modification at a time; combinations are untested.

5 Future Work

- **Combined α-scheduling.** A schedule jointly conditioned on send-progress μ and open-progress $\tilde{o}$ rather than either alone (Section 3.1); the main obstacle is the added hyperparameter count.
- **Plug-and-play composition under the factored score.** Systematically combine the primitive-level refinements — recency-weighted ϕ_j, forward-pass $f_j(t)$, variance-based p_j, TTO- or deep-autoencoder-derived Σ — inside $\pi_j(t)$ and $u_j(t)$ (Eq. (10)) rather than evaluating each in isolation,

and search jointly rather than ablating one factor at a time, since the gains in Section 4.3 are not guaranteed to be additive.

- **Hyperbolic decay for Y.** $Y_{i,j} = \kappa_\delta/(\Delta_{i,j} + \kappa_\delta)$ gives a heavier tail than the exponential label of Eq. (12) and merits a direct comparison.
- **TTO-to-TTO autoencoder.** Replacing the SAE input with Y as well, $\min_{E,D} \ell(Y, \sigma(YB_{E,D}))$, exposes TTO on both sides but changes the role of e_j — an instantaneous open becomes a boundary point rather than a typical input — so p_j can no longer be queried at e_j and would need re-deriving, e.g. via a recipient- or population-level mean-TTO query, with unclear risk of double-counting against ϕ_j.
- **Asymmetric deep encoder/decoder.** The two-layer network of Section 3.7 uses matched encoder/decoder depth and width; decoupling them (e.g. a wider encoder with a narrower, more constrained decoder, or vice versa) may better balance representational capacity against the denoising-based regularization, and warrants exploration alongside a wider sweep over masking rate, dropout, and bottleneck width d to address the variance observed in Section 4.3.

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